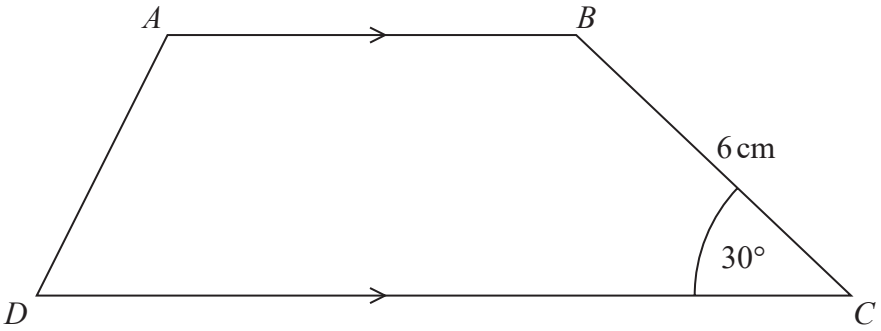
**18** Here is trapezium $ABCD$.




The area of the trapezium is $66\,cm^2$

the length of $AB$ : the length of $CD = 2 : 3$

Find the length of $AB$.

Height h = 6 sin 30° = 6 × ½ = 3 cm

Area = ½(AB + CD)h = 66

Therefore AB + CD = 44 cm

AB : CD = 2 : 3, so 5 parts = 44

AB = (2/5) × 44 = 17.6 cm

**1 mark**
sin 30° = h/6

**1 mark**
h = 3 cm

**1 mark**
trapezium area formula

**1 mark**
ratio gives 2/5 of 44

**1 mark**
AB = 17.6 cm

17.6 cm

**(Total for Question 18 is 5 marks)**

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA